Oral presentation and report

The grade for this course will be based on a group project that results in an oral presentation in which every group member presents, and a collaboratively written report. The project involves building a model to fit data, and evaluating that model. As discussed in lecture, the goal is to go in depth on at least one part of the overall Bayesian workflow presented in the course, and show how it fits within the broader workflow. The primary focus could be on any part of that process: the model itself and how you build it, the method you use to fit it, or the methods you use to verify your result. Potential topics are shared on the course Moodle page.

4 categories × 25 points each for the presentation and for the report.

Final grade: The grades of the oral presentation and written report get averaged to produce one score. The final mark is a linear map from [40,80] to [4,6]. Anything greater than 80 will receive a 6.

In each category: insufficient = 0–9 points; developing = 10–16; competent = 17–20; exemplary = 21–25. “Exemplary” is what I would expect of myself, Uddipta or Giada, so you can already get full points for “competent.”

Group sizes: People must work in groups of 2–4. If you do not have a group, check the sign-up spreadsheet or speak with one of us.

Oral presentation: Categories 1, 2, 3 are graded as a group while category 4 is graded individually. The oral presentations will be scheduled for the last 2 or 3 weeks of the course.

Written report: One report for the whole group will be turned in. There will be one grade given for the whole group on the written report. Please include a paragraph (“contribution statement”) in which group members briefly detail the parts of the analysis they performed. Names and LEGI numbers should be clearly written at the top of the report.

The written report should be a **maximum of 5 pages (excluding references and contribution statement)**. The report should be written in the style of a journal article, with typeset equations (preferably LaTeX) and references. I will include an example TeX project that people may use. Online platforms such as Overleaf are very nice for these purposes and make it very easy to collaborate with colleagues.

The written report will be due **Friday, January 15th**, and submitted through the Moodle portal.

Git repository: Along with the written report, groups must provide **a link to an associated git repository** with scripts and notebooks that produce the plots that are in the report. *If no git repository is provided, or the repository doesn’t clearly show the work that went into the report, it will mean a loss of 10 points from the averaged score (before conversion to the final mark).*

Rubric

1. Understanding of the material

Assesses: Understanding of the statistical material presented, particularly the statistical method or model that is the main focus of the report and presentation.

Exemplary: Shows a command of the material; explains the relevant method/model and background information accurately and in depth; shows an understanding of subtleties and limitations of the method/model.

Competent: Shows an understanding of core ideas; there may be some minor inaccuracies or some nuance missing.

Developing: Shows a partial understanding; key points of the method/model are missing or unclear.

Insufficient: Major misunderstandings or lack of comprehension.

2. Context and synthesis within a general Bayesian workflow

Assesses: Ability to situate the main method/model within the broader Bayesian data analysis workflow presented in class, and relate the problem to the broader topic of Bayesian data analysis.

Exemplary: Clearly articulates how the method/model solves a problem of interest, why it is needed, and how it fits within a well-developed Bayesian workflow.

Competent: Includes some context and motivation for the method/model; how it fits into a Bayesian workflow is discussed but maybe not fully developed.

Developing: Mentions context and motivation superficially; limited integration into the text and passing or shallow mention of how it fits into a full Bayesian workflow.

Insufficient: No meaningful context or synthesis.

3. Critical analysis and insight

Assesses: Ability to go beyond summary—identify strengths, weaknesses, assumptions, limitations, and potential future directions.

Exemplary: Offers thoughtful critique; recognizes assumptions; suggests insightful extensions or implications.

Competent: Some critique; acknowledges limitations but lacks depth.

Developing: Minimal critical analysis; mostly descriptive.

Insufficient: No critique or incorrect/incoherent evaluation.

4 (oral presentation only). Clarity and quality of oral presentation

Assesses: Organization, pacing, communication, slide design, correct use of figures/equations, and audience engagement.

Exemplary: Well-structured, polished, uses visuals effectively, communicates complex ideas clearly.

Competent: Mostly clear and organized; visuals adequate; some roughness.

Developing: Uneven organization or clarity; visuals cluttered or insufficient.

Insufficient: Hard to follow; poor communication or organization.

4 (report only). Quality of written report

Assesses: Accuracy, clarity, organization, conciseness, and quality of explanation in the five-page write-up.

Exemplary: Well-structured, precise, technically accurate, with clear writing and insightful summary.

Competent: Understandable and mostly accurate; minor structural or clarity issues.

Developing: Writing unclear or disorganized; multiple errors or omissions.

Insufficient: Incomplete, confusing, or highly inaccurate.